

Controlled Lyric Generation

Controlled Lyric Generation with Mood and Decade Conditioning

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DATA446 - Generative AI

Spring 2026

Abstract

This project studies controlled lyric generation using a pre-trained causal language model. The goal is to generate short lyric samples that are conditioned on user-specified attributes, which in this project includes mood and decade. The controlled short lyric samples are compared against an uncontrolled baseline sample. A cleaned subset of Billboard song lyrics was used to train both the controlled model and baseline model. Results show that the controlled model successfully incorporated the prompt conditioning, however, improvements over the baseline model were marginal. The findings suggest that controlled lyric generation is workable, though performance is limited due to the dataset size and the difficulty of learning the stylistic lyric structure from a smaller lyric collection.

Introduction

The objective of this project is to design a generative AI system that can produce short song lyrics based on attributes that users can input such as the mood and decade. Rather than generating random text lyrics, the goal is to create lyrics that better match the user’s desired musical direction. Generated random lyrics should be based in a particular style; otherwise, a songwriter who is writing an upbeat pop song may receive generated lyrics that fit a slower, ballad-like song. This project will study the problem of how to make generative models more useful for creative assistance by improving their stylistic consistency.

This project focuses on the problem of controlled lyric generation. Rather than generating unrestricted lyrics, the goal is to generate short lyric samples conditioned on two user controls: mood and decade. In this setting, a user may request lyrics that sound “sad” and “1990s”, or “happy” and “2000s.” The core research question is whether adding such tokens to the model input will lead to more useful and stylistic lyric outputs than a baseline model without the control tokens.

The project was implemented using a pre-trained causal model, DistilGPT-2. A cleaned subset of Billboard lyric data was then split by song into training, validation, and test sets resulting in 167 training examples, 36 validation examples, and 36 test examples. This split was done on a song level to avoid leakage across divisions.

Related Work

Recent work has explored lyric generation using deep learning models trained on existing song text. Dhandapani et al. (2023) looks into song lyric generation using LSTM and RNN-based models, which focus on generating lyrics for a specific genre and artist from lyrics. Their model

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uses a multilayer LSTM architecture and could generate both pop and rap lyrics that closely resemble properties such as length and variation in the training data. This work is relevant because it asserts that neural language models can learn from stylistic patterns of lyrics to generate lyrics. However, this approach emphasizes text similarity rather than direct user control over musical attributes such as mood. My project builds on the general idea of neural lyric generation, but the focus is more specifically on controlled lyric generation, where users can control the output using mood and decade attributes, and evaluation will have stylistic usefulness for songwriting support.

Method

This project will have a controllable lyric generation system built on top of a pre-trained causal model. The main idea is to take structured control tokens (mood and decade) to lyric prompts during training. This will be designed on top of prior controllable generation work to show that control codes can help model outputs towards a desired style. The methodology falls under an LLM-style autoregressive generation, with fine-tuning a pre-trained causal transformer on a domain-specific dataset. DistilGPT-2 was used because it is much smaller and more manageable to fine-tune than a full GPT-2, while still supporting open-ended next-token generation.

Dataset Construction

The original source was based on Billboard popular song lyrics. To make the data more manageable, a smaller, cleaned dataset subset was created. Song lyrics were cleaned, split into smaller lyric chunks, and labeled with attributes such as mood and decade. The decade labels

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were taken from the song release year, while mood labels were assigned manually under a specified label guide. After the cleaning and labeling, the data was split by unique song into training, validation, and test sets. The final split contained 167 training examples, 36 validation examples, and 36 test examples.

Text Formatting

Two model variants were trained in this project. The baseline model used only lyric text as training input. Its objective was to learn general lyrical continuation patterns without any explicit stylistic guidance. The controlled model used the same lyric text, but each training example was prefixed with control tokens representing mood and decade. For example, a training item might begin with a prompt such as: <MOOD=sad> <DECADE=1990s>, followed by the lyric chunk. This formatting helped the model associate specific lyric patterns with these attribute tokens. During generation, only the control prompt was provided and the model was asked to continue on from it.

Training Procedure

Both models were fine-tuned using the Hugging Face Transformers framework. The text was tokenized and then grouped into specific blocks of 128 tokens. These were then used for the standard causal language modeling, where the model will predict the next token in the sequence. During training, the controlled model showed validation loss decreasing from 3.8295 to 3.7900. This indicates that the model was able to learn from the conditioned text format.

Technical Challenges

The main technical challenge was that lyric data is both small and very repetitive. Various songs have similar lines of lyrics and have common phrases, which makes the language model vulnerable to repetition in loops and generate generic outputs. Another challenge is that the lyric

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style is not clearly encoded by one or two labels, with mood and decade as significant controls, but only capturing a part of what makes lyrics sound stylistically unique. Lastly, the cleaned final dataset used was relatively small, so the controlled model had a limited amount of evidence to learn a strong association between the labels and outputs.

Novel Ideas

The main idea in this methodology is to adapt a pre-trained causal language model to lyrical text, and test whether fine prompt conditioning improves usefulness compared with an uncontrolled baseline model. This is not a new architecture, but a controlled generation composition for lyric assistance using “mood” and “decade” tokens.

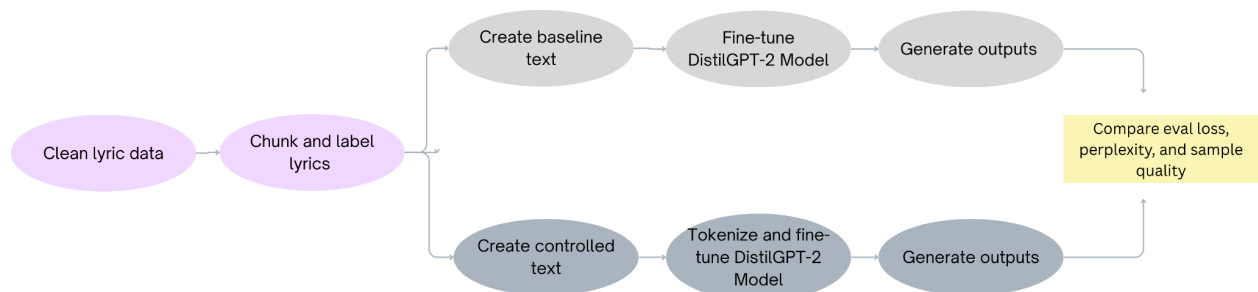


Figure 1.

Controlled Lyric Generation Workflow

Results

The experiment compared two models: one uncontrolled baseline model and one controlled model conditioned on mood and decade attributes. Both models were trained on the

same cleaned lyric subset and evaluated on validation and test data. In the final evaluation, the baseline model achieved an evaluation loss of 3.7981 and a perplexity of 44.62, whereas the controlled model obtained an evaluation loss of 3.7863 and a perplexity of 44.09. Since lower evaluation loss and lower perplexity indicate better language modeling performance, these results suggest that the controlled model outperformed the baseline on standard evaluation metrics, but the margin was minimal.

Both models are capable of producing lyric-like text, but both also struggled with coherence. The baseline model often generated generic and repetitive lyrics, while the controlled model produced the requested prompt attributes intermittently. The prompt tokens in the controlled samples were preserved and the model recognized them, however, the actual lyric content had marginal differences.

baseline_prompt	baseline_output	controlled_prompt	controlled_output
write short song lyrics	write short song lyrics by the way i was just a little boy with no passion for love and so when you said that my heart couldnt be filled but what your soul did is make me cry now to know how long will it take before we can find someone like myself who loves us better than yours truly loved by all	<MOOD=happy> <DECADE=1980s>	<MOOD=happy> <DECADE=1980s> theres been a long day and now i want to show you that weve got the right things here but im tired of being alone so let me tell ya about how much love your body makes for an interesting thing with all my friends on earth who are just as successful at it this time in our
write short song lyrics	write short song lyrics the sound i was doing is so strong it keeps you warm all around the room and when my heart breaks keep me cold stay close to a fire till midnight always remember that dream cause its true baby girl she loves her music but oh yeahits not just about love or romance im in high school with one	<MOOD=sad> <DECADE=1990s>	<MOOD=sad> <DECADE=1990s> i came to love my mom but i cant take it away from you and now this is what a sweet day we were all used for the dream come true that she told me how beautiful our hearts are so filled with joys like happiness when her heart breaks down your tears cause why can't they be
write short song lyrics	write short song lyrics dont want to make your own music but you know that theres nothing like it then its not gonna happen if i dont do what im supposedto be so bad now how about something better and more right here in my heart just wanna say hello oh yeah weve got all this together for a while	<MOOD=reflective> <DECADE=1970s>	<MOOD=reflective> <DECADE=1970s> bam i was really lost in the music of a million years ago now im just going mad at you all for what they say and why didnt we do it again with its your turn to make my way outta town tonight looking around id dont see that long till after this songyou can dance like
write short song lyrics	write short song lyrics what i wanna hear all the time but we can't play without you and when love hits me oh yeah cause that is why im so glad to see this little girl be loved like a star in your life just for her amazing voice shes my heroi cant deny it who they arebut hey now	<MOOD=confident> <DECADE=2000s>	<MOOD=confident> <DECADE=2000s> i got the feeling that i was in denial because of how you and my family have been together for over a century now im on so bad its just going to get better every time your face comes up at me cause they always say it hurts but when someone tells them thats why noone is thereand

Table 1.

Baseline vs. Controlled Sample Outputs

These results suggest that the controlled approach was partially successful, where the model learned the prompt format and could generate continuations based from the conditioned inputs. A likely explanation for the lack of performance in the controlled model is the limited size on the dataset. Another is that the attributes of “mood” and “decade” are relatively vague labels, and lyrical styles may depend on more indirect labels such as theme, structure, and song structure patterns. Overall, this project’s results are limited but informative, showing that controlled lyric generation can be implemented using a small pre-trained transformer and manually labeled collections.

Discussion

This project explored whether mood and decade conditioned lyric generation could improve over an uncontrolled baseline. The controlled model successfully learned the prompt format and generated lyric-like continuations, but the improvements over the baseline were minimal. The main takeaway is that lyric generation is attainable, but limited by small datasets and modest labels. Future research could use larger datasets, add more enhanced labels, and test stronger evaluation methods for lyrical creativity and stylistic alignment.

References

- Dhandapani, A., Ilakiyaselvan, N., Mandal, S., Bhadra, S., & Viswanathan, V. (2023). *Lyrics Generation Using LSTM and RNN*. In *Big Data and Cloud Computing*. Springer.
- Monteith, K., Martinez, T., & Ventura, D. (2012). *Automatic Generation of Melodic Accompaniments for Lyrics*. In *Proceedings of the International Conference on Computational Creativity*.